

POLYNOMIAL division by a linear polynomial lies at the heart of many results in algebra. Beginning with the proof of the polynomial remainder theorem, this note explores the nature of roots, factors, and reducibility of polynomials. It then shows that the ability to factor a polynomial depends on the choice of coefficient domain and how passing to extension fields uncovers additional roots. The concluding example shows these ideas in action and introduces the concepts of splitting fields and root symmetries, providing a first glimpse of Galois theory.

THEOREM. *In an integral domain A , for every polynomial P in the variable x with coefficients in A , $P \in A[x]$, and every $r \in A$, there exist unique $Q \in A[x]$ and $R \in A$ such that*

$$P(x) = (x - r)Q(x) + R,$$

where

$$R = P(r).$$

○ 1. The existence

Let $P(x)$ be a polynomial in $A[x]$ of degree n ,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i,$$

where each coefficient $a_i \in A$. Evaluating it at $x = r$ gives

$$P(r) = a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = \sum_{i=0}^n a_i r^i.$$

Consider the difference

$$P(x) - P(r) = \sum_{i=0}^n a_i x^i - \sum_{i=0}^n a_i r^i.$$

By the linearity of summation, the two expressions can be combined term-by-term. Moreover, the constant terms for $i=0$ cancel out ($x^0 = r^0 = 1 \Rightarrow a_0 x^0 - a_0 r^0 = a_0 - a_0 = 0$). The difference $P(x) - P(r)$ becomes

$$\begin{aligned} \sum_{i=0}^n a_i x^i - \sum_{i=0}^n a_i r^i &= (a_0 - a_0) + \sum_{i=1}^n (a_i x^i - a_i r^i) \\ &= \sum_{i=1}^n a_i (x^i - r^i). \end{aligned}$$

Each term $x^i - r^i$ in the above sum is a difference of two i -th powers. The following lemma on the factorization of $x^n - c^n$ for any $n \geq 1$ shows that every such difference can be factored by $(x - r)$. Applying the lemma term-by-term exposes a common factor $(x - r)$ across all the terms, so it can be factored out of the entire expression.

LEMMA. For any integer $n \geq 1$, the difference of two n -th powers, $x^n - c^n$, can be factored by $(x - c)$. That is, there exists a polynomial $S_{n-1}(x, c)$ such that

$$x^n - c^n = (x - c)S_{n-1}(x, c).$$

○ (by mathematical induction on n)

The **base case** $n = 1$ holds since

$$x^1 - c^1 = (x - c)(1),$$

where the constant $1 = S_0(x, c)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $S_{k-1}(x, c)$ such that

$$x^k - c^k = (x - c)S_{k-1}(x, c).$$

Induction Step. To show that the lemma holds for $k + 1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^kx$ and grouping as

$$x^{k+1} - xc^k + c^kx - c^{k+1} = (x^{k+1} - xc^k) + (c^kx - c^{k+1}).$$

Factoring out the common components from each group gives

$$x(x^k - c^k) + c^k(x - c).$$

Applying the inductive hypothesis to substitute $(x - c)S_{k-1}(x, c)$ for $(x^k - c^k)$ yields

$$x(x - c)S_{k-1}(x, c) + c^k(x - c).$$

Factoring out the common multiplier $(x - c)$ results in

$$(x - c)(xS_{k-1}(x, c) + c^k),$$

where $xS_{k-1}(x, c) + c^k = S_k(x, c)$ is a polynomial — since $S_{k-1}(x, c)$ is a polynomial by the inductive hypothesis, and polynomial rings are closed under multiplication and addition.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $S(x, c)$ polynomials are

$$\begin{aligned}
1 &= S_0 \\
x(1) + c^1 &= x + c &= S_1 \\
x(x + c) + c^2 &= x^2 + xc + c^2 &= S_2 \\
x(x^2 + xc + c^2) + c^3 &= x^3 + x^2c + xc^2 + c^3 &= S_3 \\
x(x^3 + x^2c + xc^2 + c^3) + c^4 &= x^4 + x^3c + x^2c^2 + xc^3 + c^4 &= S_4 \\
&\dots \\
x(x^{n-2} + x^{n-3}c + \dots + xc^{n-3} + c^{n-2}) + c^{n-1} &= x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1} = S_{n-1} \\
x(x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1}) + c^n &= x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = S_n
\end{aligned}$$

In more compact notation with the summation symbol

$$S_n(x, c) = x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = \sum_{i=0}^n x^{n-i}c^i$$

and

$$xS_{n-1}(x, c) + c^n = x \sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^n x^{n-i}c^i = S_n(x, c).$$

The exact factorization is $x^n - c^n = (x - c) \sum_{i=0}^{n-1} x^{n-1-i}c^i$.

Several interesting properties characterize $S_n(x, c) = \sum_{i=0}^n x^{n-i}c^i$ as a *complete homogeneous symmetric polynomial* in two variables :

- ✓ it is *symmetric*, since $S_n(x, c) = \sum_{i=0}^n x^{n-i}c^i = \sum_{i=0}^n c^{n-i}x^i = S_n(c, x)$,
- ✓ it is *homogeneous* of degree n , because the degree of each monomial $x^{n-i}c^i$ is exactly $n-i+i=n$ for all i ,
- ✓ it is *complete* in the sense that it contains every possible n -th degree monomial in the variables x and c exactly once, each with a coefficient of 1.

The symmetry of $S_n(x, c)$ is already hinted at in the induction step. The proof inserts the zero $0 = -xc^k + c^kx$, but one may equally insert the zero $0 = -x^kc + cx^k$. Then $x^{k+1} - x^kc + cx^k - c^{k+1} = x^k(x - c) + c(x^k - c^k)$ and the resulting recurrence will be

$$S_k(x, c) = x^k + cS_{k-1}(x, c).$$

The recurrences $x^k + cS_{k-1}(x, c)$ and $xS_{k-1}(x, c) + c^k$ are dual under the swap $x \leftrightarrow c$.

Returning to
$$P(x) - P(r) = \sum_{i=1}^n a_i(x^i - r^i),$$

the lemma yields*

$$x^i - r^i = (x - r)S_{i-1}(x, r)$$

for every $i \geq 1$.

* From a deeper algebraic viewpoint, $S_{i-1}(x, c)$ from the lemma belongs to the polynomial ring $\mathbb{A}[x, c]$. Replacing the indeterminate c by the element $r \in \mathbb{A}$ is the evaluation homomorphism $\mathbb{A}[x, c] \rightarrow \mathbb{A}[x]$. Thus $S_{i-1}(x, r)$ is a polynomial in $\mathbb{A}[x]$.

Substituting these factorizations term-by-term gives

$$P(x) - P(r) = \sum_{i=1}^n a_i (x - r) S_{i-1}(x, r) = (x - r) \sum_{i=1}^n a_i S_{i-1}(x, r).$$

Since each $S_{i-1}(x, r)$ is a polynomial in $A[x]$ and polynomial sums are polynomials, the sum $\sum_{i=1}^n a_i S_{i-1}(x, r)$ belongs to the same polynomial ring $A[x]$. Define

$$\sum_{i=1}^n a_i S_{i-1}(x, r) = Q(x) \in A[x].$$

$$\text{Hence} \quad P(x) - P(r) = (x - r) Q(x)$$

$$\text{or equivalently} \quad P(x) = (x - r) Q(x) + P(r)$$

— the latter exhibits the intended representation with $\mathcal{R} = P(r) \in A$. Ergo, such $Q \in A[x]$ and $\mathcal{R} \in A$ exist. ●

It remains to show that $Q(x)$ and $\mathcal{R}$ are unique.

○ 2. The uniqueness

Assume that there exist polynomials $Q', Q'' \in A[x]$ and constants $R', R'' \in A$ such that

$$P(x) = (x - r) Q'(x) + R' \quad \text{and} \quad P(x) = (x - r) Q''(x) + R''.$$

Subtracting the second representation from the first gives

$$(x - r) Q'(x) + R' - (x - r) Q''(x) - R'' = 0,$$

which simplifies to

$$(x - r)(Q'(x) - Q''(x)) = R'' - R'.$$

Evaluating this polynomial identity at $x = r$, the factor $(x - r)$ becomes $r - r = 0$, so the left-hand side vanishes and reveals* that the constant difference is zero :

$$0 = R'' - R',$$

that is

$$R' = R''.$$

* Since $R', R'' \in A$, the difference $R'' - R'$ is an element of A itself — equivalently, a constant polynomial in $A[x]$. Accordingly, once evaluation at $x = r$ yields $0 = R'' - R'$, it follows that $R' = R''$ as elements of A . In other words, $R' = R''$ is an equality in A itself, not merely equality obtained after evaluating the polynomial identity at $x = r$.

Returning to the polynomial identity before evaluation and putting $R' = R''$ into it yields

$$(x - r)(Q'(x) - Q''(x)) = 0.$$

Because A is an integral domain, the polynomial ring $A[x]$ is an integral domain as well and therefore has no zero divisors. Moreover, $x - r$ is a nonzero polynomial in $A[x]$. Since its product with $Q'(x) - Q''(x)$ is the zero polynomial, the absence of zero divisors in $A[x]$ implies that

$$Q'(x) - Q''(x) = 0,$$

giving

$$Q'(x) = Q''(x).$$

Thus
$$Q' = Q'' \quad \text{and} \quad R' = R'',$$

showing that the intended representation

$$P(x) = (x - r)Q(x) + R$$

is unique in the choice of Q and R . ●

DEFINITION. Let $P \in A[x]$ and $r \in A$. If $P(r) = 0$, then r is called a **root** (or **zero**) of the polynomial P . Equivalently, r is a solution of the polynomial equation $P(x) = 0$.

COROLLARY. If $r \in A$ is a root (solution) of the polynomial equation $P(x) = 0$, then there exists unique $Q \in A[x]$ such that

$$P(x) = (x - r)Q(x).$$

○ By the previous theorem, there exists unique $Q \in A[x]$ such that

$$P(x) = (x - r)Q(x) + P(r).$$

Since r is a root, $P(r) = 0$. Therefore

$$P(x) = (x - r)Q(x). \quad \bullet$$

In this case, $(x - r)$ is called a *factor* of $P(x)$, just as one says that a number b is a factor of another number a when b divides a . Hence every root $r \in A$ gives a linear factor $(x - r)$ of the polynomial.

The above definition concerns roots that belong to the same coefficient domain A . However, a polynomial in $A[x]$ may fail to have roots inside A and acquire them only in a larger domain or field containing A . Whenever a root r exists in such an extension, the same theorem gives factorization $P(x) = (x - r)Q(x)$ in the polynomial ring over that larger domain.

DEFINITION. A nonconstant polynomial $P \in A[x]$ is called **reducible over A** if there exist nonconstant polynomials $A(x), B(x) \in A[x]$ such that

$$P(x) = A(x)B(x).$$

If no such factorization exists, then P is called **irreducible over A** .

REMARK. Reducibility is always relative to the chosen coefficient domain. A polynomial may be irreducible over one domain and reducible over a larger one.

❖ For example, the polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$. Since it is quadratic, any nontrivial factorization would have to be into two linear factors. But it has no rational root, hence no linear factor in $\mathbb{Q}[x]$. However, it has the roots $\pm\sqrt{2}$ in the real numbers, and over $\mathbb{R}$ the same polynomial becomes reducible*

$$x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2}).$$

Thus, a factorization that is impossible over one coefficient domain may become possible after passing to a larger domain containing the roots.

❖ In contrast, the polynomial $2x - 1$ is irreducible over $\mathbb{Z}$. Yet it has the rational root $1/2$, therefore over $\mathbb{Q}$ it factors as

$$2x - 1 = \left(x - \frac{1}{2}\right)(2),$$

but the second factor is a constant polynomial. Over $\mathbb{Q}$, $2x - 1$ remains irreducible, because it cannot be written as a product of two nonconstant polynomials. Here, enlarging the coefficient domain uncovers the root, but doesn't produce a nonconstant factorization, so the polynomial can't become reducible.

Every nonconstant linear polynomial has degree 1 and is irreducible over any integral domain. Thus, only polynomials of degree two or higher can become reducible by acquiring a linear factor through one of their roots.

* Although $x^2 - 2$ splits into linear factors over the reals, the field $\mathbb{R}$ is not the *smallest* extension of $\mathbb{Q}$ in which this factorization becomes possible. A smaller field already suffices, namely $\mathbb{Q}(\sqrt{2})$. This will be explained after the concluding example.

COROLLARY. *If a polynomial $P \in \mathbf{A}[x]$ has a root $r \in \mathbf{A}$ and the degree of P is at least 2, then P is reducible over $\mathbf{A}$.*

○ If $r \in \mathbf{A}$ is a root of $P(x) = 0$, then by the first corollary there exists $Q \in \mathbf{A}[x]$ such that

$$P(x) = (x - r)Q(x).$$

Since the degree of P is at least 2, both $x - r$ and the polynomial Q are nonconstant. That is, P is a product of two nonconstant polynomials in $\mathbf{A}[x]$, therefore it is reducible over $\mathbf{A}$. ●

REMARK. The existence of a root in $\mathbf{A}$ implies reducibility over $\mathbf{A}$, but reducibility does not necessarily mean that a polynomial has a root in $\mathbf{A}$. For example, over $\mathbb{R}$ the polynomial

$$x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2)$$

is reducible, although it has no real root.

EXAMPLE *of how to factor a polynomial into all of its linear factors via sequential domain extension.*

This example explores how the factorization and reducibility of a polynomial depend on the chosen coefficient domain. As the domain is enlarged, the polynomial acquires new roots, and each newly obtained root translates into a new linear factor.

Consider the polynomial

$$P(x) = 2x^3 - x^2 - 4x + 2,$$

whose coefficients all belong to $\mathbb{Z}$. Thus initially $P \in \mathbb{Z}[x]$.

1. Over the integers $\mathbb{Z}$

By grouping the terms and applying the distributive property,

$$2x^3 - x^2 - 4x + 2 = 2x(x^2 - 2) - (x^2 - 2) = (2x - 1)(x^2 - 2).$$

While the factor $(2x - 1)$ suggests $x = 1/2$ as a candidate root and $(x^2 - 2)$ becomes zero at $x = \pm\sqrt{2}$, none of these values belong to $\mathbb{Z}$. Hence the polynomial $P(x)$ is reducible over $\mathbb{Z}$ yet has no roots in $\mathbb{Z}$.

2. Enlarging the domain from $\mathbb{Z}$ to $\mathbb{Q}$

Now regard the same polynomial as an element of $\mathbb{Q}[x]$, $\mathbb{Z} \subset \mathbb{Q}$. Then

$$P(1/2) = 0,$$

thus $1/2$ is a root in $\mathbb{Q}$. By the first corollary, there exists a polynomial $Q \in \mathbb{Q}[x]$ such that

$$P(x) = \left(x - \frac{1}{2}\right)Q(x).$$

Indeed,

$$2x - 1 = \left(x - \frac{1}{2}\right)(2).$$

Thereby, after extending the coefficient domain to $\mathbb{Q}$, the polynomial

$$P(x) = (2x - 1)(x^2 - 2) = 2\left(x - \frac{1}{2}\right)(x^2 - 2)$$

remains reducible over $\mathbb{Q}$. The first enlargement of the coefficient domain doesn't change the reducibility of P , as it was already reducible over $\mathbb{Z}$. It merely replaces the linear factor $(2x - 1)$ by its *monic associate* $(x - 1/2)$, while the quadratic factor remains irreducible over $\mathbb{Q}$.

3. Enlarging the domain further from $\mathbb{Q}$ to $\mathbb{R}$

In $\mathbb{Q}$, the remaining factor $(x^2 - 2)$ still lacks roots because $\pm\sqrt{2} \notin \mathbb{Q}$. However, in $\mathbb{R}$ it splits as

$$x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2}).$$

Therefore, in $\mathbb{R}[x]$

$$P(x) = 2\left(x - \frac{1}{2}\right)(x - \sqrt{2})(x + \sqrt{2}).$$

After the second enlargement of the coefficient domain, every irreducible factor is linear. The polynomial now splits completely into linear factors, and all of its roots become visible :

$$\frac{1}{2}, \quad \sqrt{2}, \quad -\sqrt{2}.$$

The second extension above, from $\mathbb{Q}$ to $\mathbb{R}$, is not the smallest possible one. To factor the remaining quadratic term $x^2 - 2$, it already suffices to adjoin the single irrational number $\sqrt{2}$ to $\mathbb{Q}$. This produces the field

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}.$$

Every element of this field has the form $a + b\sqrt{2}$ with rational coefficients a, b . It is an extension of $\mathbb{Q}$ since it contains all the original rational numbers $\mathbb{Q}$ when $b = 0$. Moreover, it is closed under the four arithmetic operations — anyone can add, subtract, multiply and divide elements of $\mathbb{Q}(\sqrt{2})$ and always get another number from $\mathbb{Q}(\sqrt{2})$ that fits the same $a + b\sqrt{2}$ pattern :

$$\begin{aligned} (a + b\sqrt{2}) + (c + d\sqrt{2}) &= (a + c) + (b + d)\sqrt{2}, \\ (a + b\sqrt{2}) - (c + d\sqrt{2}) &= (a - c) + (b - d)\sqrt{2}, \\ (a + b\sqrt{2})(c + d\sqrt{2}) &= (ac + 2bd) + (ad + bc)\sqrt{2}, \end{aligned}$$

division works too, because the denominator can always be “rationalized” by multiplying by the conjugate $a - b\sqrt{2}$

$$\begin{aligned}\frac{1}{a + b\sqrt{2}} &= \frac{a - b\sqrt{2}}{(a + b\sqrt{2})(a - b\sqrt{2})} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} \\ &= \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2},\end{aligned}$$

provided $a + b\sqrt{2} \neq 0$, that is a and b are not both zero.

Since $\mathbb{Q}(\sqrt{2})$ is a field containing $\mathbb{Q}$ and the roots $\pm\sqrt{2}$ of $x^2 - 2$, the polynomial from the example splits completely already in $\mathbb{Q}(\sqrt{2})[x]$,

$$P(x) = 2\left(x - \frac{1}{2}\right)(x - \sqrt{2})(x + \sqrt{2}).$$

The field $\mathbb{Q}(\sqrt{2})$ also carries a natural symmetry and can equally well be written as $\mathbb{Q}(-\sqrt{2})$, where $\sqrt{2}$ and $-\sqrt{2}$ can be swapped without changing the field itself. Under this interchange, the two linear factors $(x - \sqrt{2})$ and $(x + \sqrt{2})$ are exchanged, while their product $(x + \sqrt{2})(x - \sqrt{2}) = x^2 - 2$ remains the same. Thus the two roots play completely symmetric roles inside the splitting field.

What emerges here is a core idea of advanced algebra — rather than enlarging the coefficient domain arbitrarily, one often seeks the *smallest* field extension in which a polynomial splits into linear factors. Such a minimal field, called a *splitting field*, contains exactly the roots and the arithmetic combinations generated by them, making the intrinsic symmetries among the roots apparent and explicit, free from the redundancy of a much larger ambient field.

In the present example, the splitting field of $x^2 - 2$ over $\mathbb{Q}$ is precisely $\mathbb{Q}(\sqrt{2})$. Another familiar example of a splitting field extension is the passage from $\mathbb{R}$ to $\mathbb{C}$. The polynomial $x^2 + 1$ has no root in $\mathbb{R}$, but adjoining a root of it, $\sqrt{-1}$, produces the larger field $\mathbb{C} = \mathbb{R}(\sqrt{-1})$, in which $x^2 + 1$ splits into linear factors,

$$x^2 + 1 = (x - \sqrt{-1})(x + \sqrt{-1}).$$

For the cubic $x^3 - 2$, the real root is $\sqrt[3]{2} \in \mathbb{R}$. Yet, while every polynomial of degree 3 has three roots, the remaining two cannot be discovered in $\mathbb{R}$ since the real line only spans one dimension. Moving to the complex numbers, all three roots become visible in the complex plane as an equilateral triangle. However, the field $\mathbb{C}$ is vastly redundant. It embodies many extraneous elements and algebraic relations that have nothing to do with the algebra of $x^3 - 2$. The smallest field embracing exactly what’s needed here is the splitting field $\mathbb{Q}(\sqrt[3]{2}, \omega)$, where ω is a root of $x^3 - 1 = (x - 1)(x^2 + x + 1)$ distinct from 1, that is a root of $\omega^2 + \omega + 1 = 0$ along with $\omega^3 = 1$. The third root is the complex conjugate of the second, $\bar{\omega} = \omega^2$, and for every integer k , $\omega^{n+3k} = \omega^n$. In this exact field, the symmetries become apparent and explicit — the triangle of roots can be rigidly rotated (cyclically permuted) by multiplying each root by ω , $\sqrt[3]{2} \mapsto \omega\sqrt[3]{2} \mapsto \omega^2\sqrt[3]{2} \mapsto \sqrt[3]{2}$, or backwards by ω^2 , and reflected by complex conjugation.

The systematic study of such splitting fields, of the symmetries that permute polynomial roots while preserving the surrounding algebraic structure, and of what those symmetries reveal about the polynomial itself, is precisely what *Galois theory* is about.